

PHASE 3 – PROJECT DESIGN PHASE

DATE	03 November 2025
TEAM ID	NM2025TMID05906
PROJECT NAME	Garage Management System
MAXIMUM MARKS	4 Marks

Solution Architecture Overview

The architecture of Garage Management System (GMS) is built using Salesforce's multi-tier cloud architecture, ensuring automation, scalability, and ease of use.

Architecture Layers:

1. User Interface Layer (Frontend)- Users access system through Salesforce Lightning Experience.- Customers submit service requests and track progress.- Service advisors manage work orders.- Mechanics update diagnostics and service tasks.
2. Application Logic Layer- Built using Apex Classes, Apex Triggers, Flows, and Validation Rules.- Automates work order creation and mechanic assignment.- Validates spare parts availability.- Sends notifications to customers.
3. Database Layer- Uses Salesforce Custom Objects: Vehicle, Customer, Service Request, Work Order, Service Task, Mechanic, Spare Parts Inventory.- Relationships ensure data integrity.
4. Reporting Layer - Reports and Dashboards display:- Daily service summary- Mechanic performance- Revenue reports- Spare parts inventory status Data Flow Summary:
 1. Customer creates Service Request.
 2. Flow/Trigger creates Work Order and assigns Service Advisor.
 3. Advisor assigns task to Mechanic.
 4. Mechanic updates progress.
5. Inventory auto-updates for parts used.
6. Admin verifies completion and billing.
7. Customer receives notifications.
8. Dashboards update in real time.

System Architecture Overview (Text Diagram):

